AI-Based Early Childhood Emotional Intelligence Monitoring Platform in Urban areas

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Project Proposal Report

IT22315250

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B.Sc. (Hons) Degree in Information Technology

Department of Computer Science and

Data Science

Sri Lanka Institute of Information Technology

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August 2025

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I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Date .....27/08/2025.....

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Date 27/08/2025

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Abstract

This research aims to identify the emotional intelligence of children aged 4–6 years, a developmental stage where complex emotions begin to emerge, making early detection vital. The focus is on recognizing emotions—an essential component of emotional intelligence—to support early intervention in emotional distress. The primary goal is to uncover hidden or suppressed emotions using indirect, culturally appropriate methods, rather than digital techniques such as facial recognition that are common in prior research.

Facial expression analysis, though widely used, has several limitations in early childhood applications. It often captures posed, static expressions that may not reflect a child's genuine emotional state. These systems are better suited for identifying extreme, exaggerated emotions, while subtle or spontaneous expressions often go undetected, reducing both accuracy and validity.

To address these issues, this study proposes using children's drawings and behavioral feedback from teachers. Drawings will be analyzed using a deep learning model trained to identify emotional categories—happy, sad, or angry—through explainable AI methods. Color clustering will be used to detect the dominant hues, and model interpretability will be enhanced by connecting visual features (e.g., color usage, drawing elements) with the predicted emotion.

Behavioral data will be collected using the Emotion Regulation Checklist (ERC), a validated tool used internationally, to cross-verify and support the model's emotional predictions.

This research will be conducted in Sri Lankan preschools. The outcome is an explainable, culturally aware system to assist teachers and parents in identifying emotional changes and hidden distress in young children, enabling timely and supportive interventions.

Keywords: Children drawings, Emotional Regulation Checklist, AI, Early childhood

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List of Abbreviations

- ERC - Emotion Regulation Checklist
- PreBERS - Preschool behavioral and emotional rating scale
- EEG – Electroencephalogram
- NVQ - National Vocational Qualification
- L/N – Lability/Negativity subscale of emotion regulation checklist
- ER – Emotion regulation subscale of emotion regulation checklist

2 Introduction

Emotional intelligence, usually described as the ability to recognize, understand, manage, and use emotions effectively in oneself and others [1]. By developing emotional intelligence, people learn more about themselves and learn to cooperate with those around them [2]. For early childhood emotional intelligence is crucial because lower emotional intelligence impacts clear thinking process and the education of the preschoolers [3].

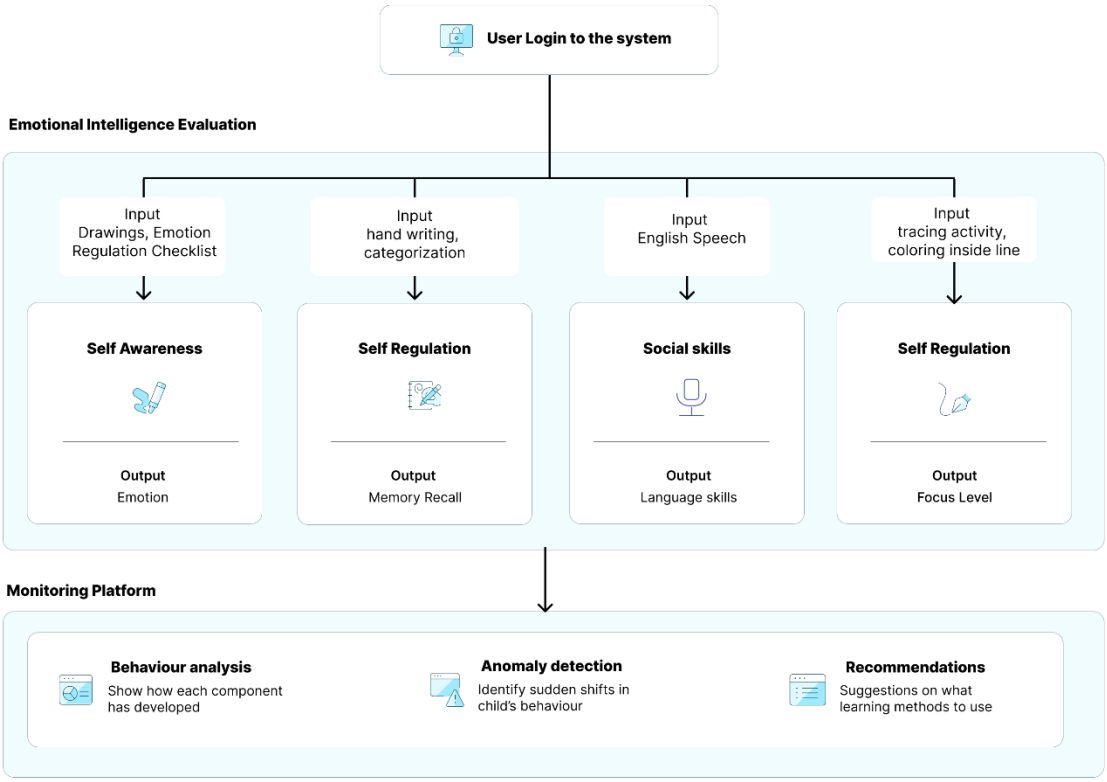


Figure 1. 1: Diagram overview

By identifying the children that are emotionally distressed earlier on, we can help them before the issue gets worse. Children maybe going through varies emotional challenges at this stage and might be hiding their true emotions. Children that are heavily impacted to these emotional changes are children who have divorced parents, children who has lost parents, children coming from violent backgrounds. From ages 4-6 children start to recognize emotions and label them correctly [4-6]. When a small child is not give the space to express their feeling, they start to lock them self and adopts an attitude of withdrawal [7]. After wards these emotions starts to become uncontrollable reactions. The three main emotions that they can be properly identified at the age of 4-6 are anger, sad and happy [4-6]. Neglecting a child's emotional needs, negatively impacts early childhood development, especially for children who cannot express their feelings

clearly[8]. The main problem the research is trying to solve is to identifying the emotions in early childhood children. The focus of this study will be children aged 4-6 in Urban areas, Sri Lanka. Prior applications that were build for the early childhood, gives the child a digital device [9-12], which increase digital addiction in children. When it comes to emotion identification, the method many of the prior researches have used is facial recognition [4].

Why facial expression applications fails?

Even though the most practiced method to identify emotions is through facial recognition methods, this methods has its limitations as well. Below are some of the limitations in facial recognition usage to identifying emotions [5],

- Facial expression applications creates a privacy issue where the child's face is monitored by the application. This has ethical because younger children are involved.
- A specific facial expression (like a frown) doesn't always and only mean a specific emotion (like sadness). The same facial expression can be associated with different emotions, and the same emotion can be expressed by different facial configurations.
- It is also invalid to just look at the facial expressions alone with out considering the background context. For example, A wide-eyed expression might indicate fear in a dangerous situation but surprise at a birthday party.
- Facial expression is trained in intense emotional states, which is not a daily life behavior.
- Facial expression datasets fail to capture the authentic emotions, the data of images are tailored onto look like the emotion. Which in turn lacks the genuine emotional state.
- Most of the facial expression application will give a task to the student and then get the emotional state from the facial expression. Depending on the complexity of the task facial expressions will change, also the complexity of the task will change from person to person.
- Another limitation facial expressions have is social Masking, where a person hide their true emotions intentionally or unintentionally [13].

Some studies have also proposed electroencephalogram (EEG) methods to identify the emotions. These produce valuable real time results. However these type of methods are uncomfortable and development of applications for regular uses are limited [13].

Benefits the user will have by using the application proposed by the research

There are multiple benefits of using methods such as Emotion Regulation Checklist (ERC) and artwork to identify emotions.

- The application helps detect emotional distress a child is going through, without waiting for the emotions to overload and make the situation worse.
- The proposed methods are non verbal ways that helps us identify the emotions. This method is ideal for children who are having difficulty in expressing their feelings in a verbal manner.
- The proposed method is an activity that can be monitored through the child's daily activities. Making it a easy to use application

3 BACKGROUND AND LITERATURE

We carried out an online survey to find out how important it is to keep track of emotional intelligence in early childhood. This survey was conducted by sending the online form to parents and the teachers in urban areas in Western province. The survey results will help in building the application in a more successful manner. The form was filled by 64 individuals. Below are the results of the form.

1. What is your role? (ඔබේ භූමිකාව කුමක්ද?)

Figure 2.1 1: Survey Result 1

2. Type of school/setting? (පාසලේ/සැකසුමේ වර්ගය?)

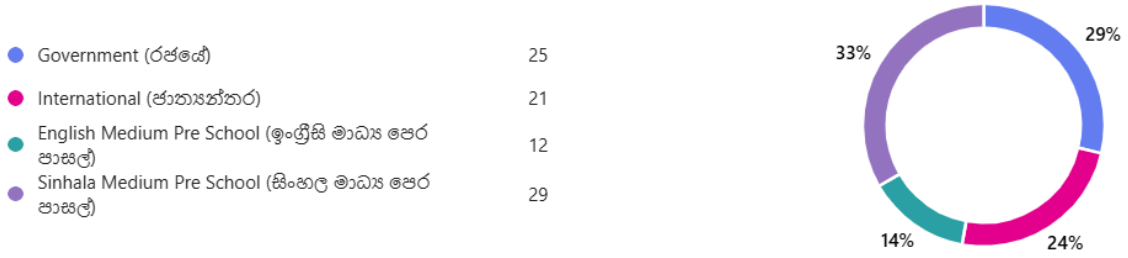


Figure 2.1 2:Survey Result 2

3. How useful would you find a tool that helps you track Memory recall, Focus, Emotion, Language Skill in your child/students? (ඔබේ දරුවා/සිසුන් තුළ මතකාවර්ජනය, අවධානය, හැඟීම්, කථන කුසලතා නිරීක්ෂණය කිරීමට ඔබට උපකාරී වන මෙවලමක් ඔබට කොතරම් ප්‍රයෝජනවත්ද?) More details

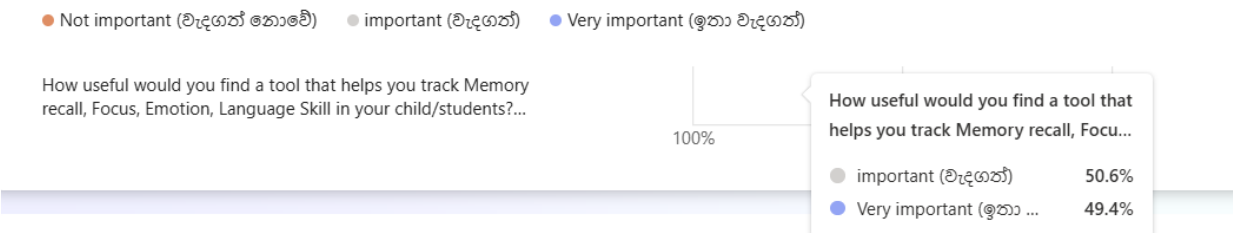


Figure 2.1 3: Survey Result 3

4. How do you currently monitor Memory recall, Focus, Emotion, Language Skills? (ඔබ දැනට මතකාවර්ජනය, අවධානය, හැඟීම්, භාෂා කුසලතා කුසලතා නිරීක්ෂණය කරන්නේ කෙසේද?) More details

Figure 2.1 4: Survey Result 4

5. Primary language used at home ? (නිවසේදී භාවිතා කරන ප්‍රාථමික භාෂාව ?) More details

Figure 2.1 5: Survey Result 5

6. Do you use English as the secondary language? (ඔබ ද්විතියික භාෂාව ලෙස ඉංග්‍රීසි භාවිතා කරනවාද?) More details

Yes	69
No	17

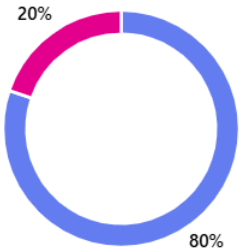


Figure 2.1 6: Survey Result 6

7. What activities does your child do most? (ඔබේ දරුවා වැඩිපුරම සිදුකරන ක්‍රියාකාරකම් මොනවාද?) More details

Writing (ලිවීම)	38
English speaking (ඉංග්‍රීසි කථා කිරීම)	36
Drawing (ඇඳීම)	73
Tracing	48
Other	10

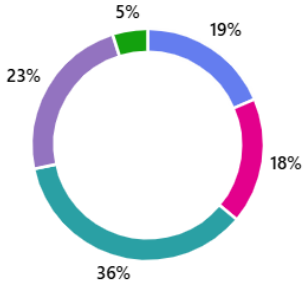


Figure 2.1 7: Survey Result 7

8. How often does your child write letters at home ? (ඔබේ දරුවා නිවසේදී කොපමණ වාරයක් අකුරු ලියනවාද ?) More details

Mostly English letters (More than 4 days a week) (බොහෝ විට ඉංග්‍රීසි අකුරු (සතියකට දින 4 කට...	12
Mostly Sinhala letters (More than 4 days a week) (බොහෝ විට සිංහල අකුරු (සතියකට දින 4 කට...	11
Both Sinhala & English letters (Less than 4 days a week) (සිංහල සහ ඉංග්‍රීසි අකුරු දෙකම (සතියක...	23
Does not write letters at home (ගෙදරදී අකුරු ලියන්නේ නෑ)	41

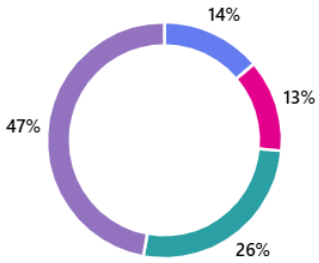


Figure 2.1 8: Survey Result 8

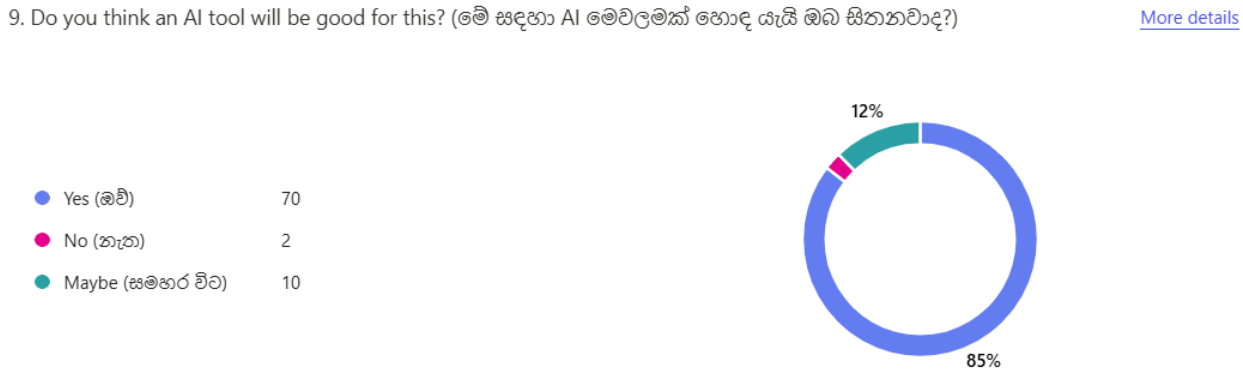


Figure 2.1 9: Survey Result 9

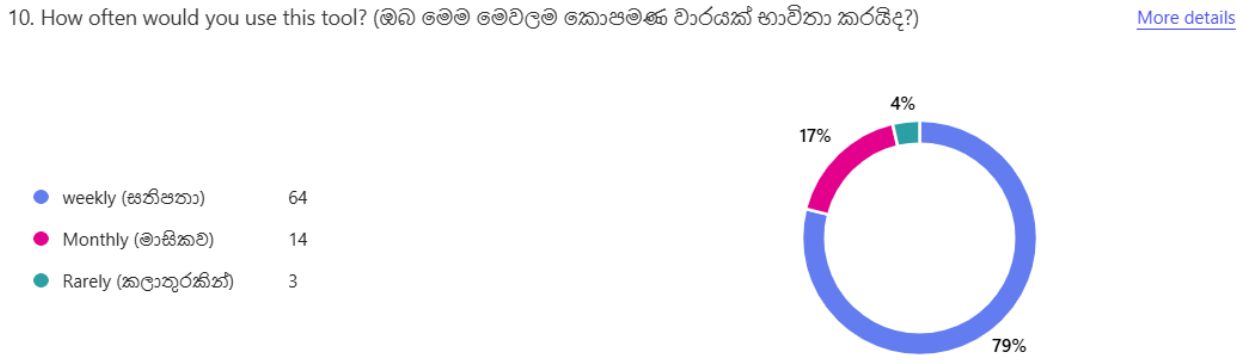


Figure 2.1 10: Survey Result 10

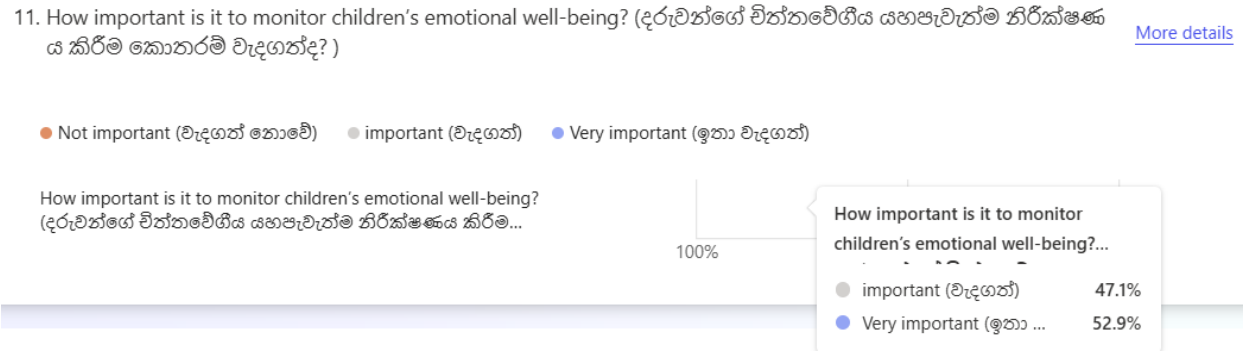


Figure 2.1 11: Survey Result 11

This survey was filled mostly by parents whose children, go to Sinhala medium preschools, mostly speaking in Sinhala and used English as the secondary language. Most of the responders agreed, a tool that can track memory recall, focus, emotion and language skill will be useful. Currently no formal method is being used for these tasks. 85% of the people agreed that an AI tool would be helpful. Out of 87 people who filled the form, 64 responded that this should be assessed in a weekly basis. Overall, every single respondent agreed that monitoring a child's emotional well being is a crucial tack.

For identification of emotions two inputs are considered by the application. Artwork and behavior metrics. These two methods complement each other and works in a way to improve the accuracy. Below is background overview of the methods.

2.1 Behavior recording metrics

2.1.2 Emotion Regulation Checklist (ERC)

Emotion Regulation Checklist contains 24 items that are rated using the scale of 1-4, where 1 = Rarely or never and 4 = Almost always. ERC has two main sub scales, emotional regulation subscale and Emotional Lability/Negativity. For this study, only the subscale of emotional lability/negativity will be used. Children with higher emotion lability/negativity show lower levels of competent social functioning [14]. Children with dysregulation show more levels of anger and aggression [14]. This checklist can be used with age groups from 3 to12 [14,15], making it a perfect scale for the emotion identification.

The Cronbach's alpha (α) coefficient for the emotional negativity/lability subscale is above 77% and in most cases it is closers to 100%, making it a reliable and valid source [14]. The results revealed an α=.66 for Emotion Regulation and an α=.80 for Lability/Negativity [14 – 16]. Factor loadings of .71 or higher can be regarded as 'excellent', 0.63 as 'very good', .55 as 'good', .45 as 'fair', and .32 as 'poor' [14].

Sub- Factor	Social Competence	Anxiety	Aggression	Withdrawing	Anger
Lability/Negativity	-.55**	.47**	.37**	.34**	.31*
Emotion Regulation	.41**	-.47**	-.32**	-.42**	-.31**

**p<.01

Figure 1.1.2 1: Relationship between Emotion Regulation Checklist and Pre-School Social Competence and Behavior Evaluation Inventory- Teacher Form [14]

2.1.2 Preschool Behavioral and Emotional Rating Scale (PreBERS)

PreBERs scale contains 4 subscales, emotional regulation, school readiness, social confidence, family involvement. This scale contains 42 items. The age category for this group is 3-5. This scale mostly access the emotional regulation [17-19].

Comparison between ERC lability/negativity and PreBERS emotional regulation subscales

ERC Lability/negativity	**PreBERs Emotional Regulation**
Emotional dysregulation and negative reactivity	Emotional regulation skills and adaptive behavior
Focus on negative affect and instability	Focus on regulation and control
Strong, widely used, validated	Emerging, less extensive
Ideal for identifying emotional problems or distress	Good for assessing regulation competence
Age 3-12	Age 3-5

Table 2.1.2.1: Comparison between ERC and PreBERS

After the above comparison the Emotion Regulation Checklist subscale of negativity/lability was considered the most appropriate tool because the high score on this reveals negative emotions (anger and sad), low scores reveal positive emotions (happy) which aligns with the results of the artwork classification. The Emotion Regulation Checklist is a widely used research tool, that has been validated in multiple countries. It is used to identify emotional distress and fits the age category of the research.

2.2 Drawings and emotions

Art is a non verbal form that can be used to express feelings, especially in children [7]. Art helps understand emotions, which is a main part of emotional intelligence [20]. Researches have pointed out that for children with limited language, mediums such as drawings, movements, music becomes the most appropriate and only way to express themselves [7]. For this reason emotions expressed through art work is an important aspect for early childhood. Art is a form of therapy were a student can express themselves freely. This also helps them to express, any past traumatic experiences that they have faced. Psychology specialist have found that drawings are specially useful when it comes to finding out individuals personality, experience and attitudes [21]. Below are studies that were done to classify art works into emotions,

- **2.1 Emotion Analysis in Children's Drawings Using Machine Learning Techniques [22]**
- **2.2 Deep learning for studying drawing behavior: A review [23]**
- **2.3 A Children's Psychological and Mental Health Detection Model by Drawing Analysis based on Computer Vision and Deep Learning [24]**
- **2.4 Cross-Linguistic Study on Affective Impression and Language for Visual Art Using Neural Speaker [25]**
- **2.5 Sentiment analysis of ethnic artworks using deep learning [26]**
- **2.6 Sentiment analysis of art and design works using deep learning [27]**

Considering the uncomfortableness of the EEG test and the accessibility limitations of it, Identifying emotions through art works becomes a more attractive method. When considering social masking limitation in facial recognition, where humans tending to hid there true emotions, the art works create more relaxed, stress free environment to express ones feelings [7]. Children at the age of 4-6 generally prefers to draw, making it a perfect input to analyze the hidden emotions.

2.2.2 Colors and emotions

When it comes to emotions, there is a connection between colors. Prior researches have categorized colors and emotion, yellow with joy, black with sadness, light colors with positive and dark colors with negative emotions [28,29]. Certain colors influences one's mood and improves well being [29,30].

When it comes to Sri Lankan preschoolers teachers are instructed to observe dark colors used by the children. Using the black color out of context in a child's drawing, has been considered as a sign of emotional distress. This knowledge is being shared to the preschool teachers by the government through seminars they conduct.

LIGHT/BRIGHT			DARK			SATURATED			DESATURATED		
Affective terms	*n*	% from total articles (*n*=7)	Affective terms	*n*	% from total articles (*n*=7)	Affective terms	*n*	% from total articles (*n*=4)	Affective terms	*n*	% from total articles (*n*=3)
Happy/joy/merry	5	71.43	Fear	4	57.14	Happy/joy	4	100.00	Sad/sadness	3	100.00
Calming/relaxation	2	28.57	Sad/sadness/ gloomy	4	57.14	Relaxation	1	25.00	Fear/fright	2	66.67
Fright	1	14.29	Angry/anger	3	42.86	Strong	1	25.00	Angry	1	33.33
Hope	1	14.29	Disgust	2	28.57				Weak	1	33.33
Love	1	14.29	Powerful/strong	2	28.57				Worried	1	33.33
Relieving	1	14.29	Worried	1	14.29						
Surprise	1	14.29									
Weak	1	14.29									

Figure 2.2.2 1:The number of articles reporting a correspondence between discrete affective terms and categories light/bright, dark, saturated, and desaturated [29]

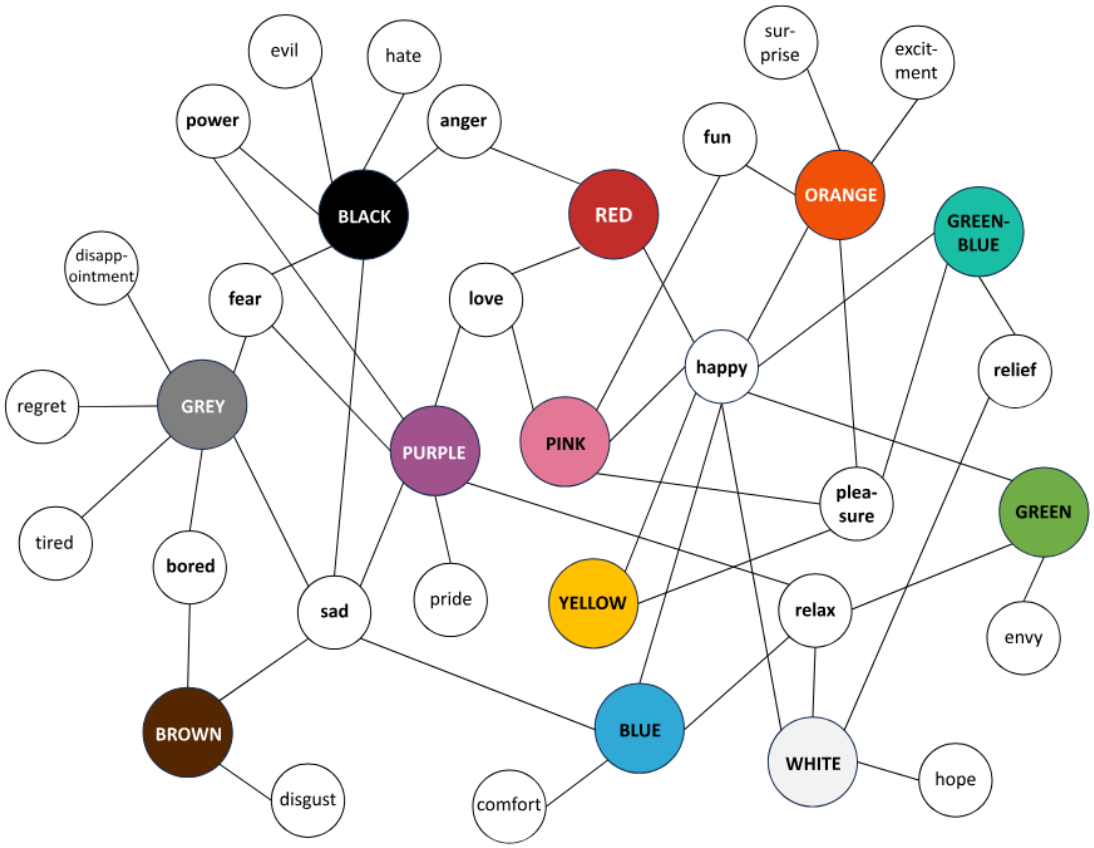


Figure 2.2.2 2: A visual representation of the most frequent correspondences between affective concepts and 12 colour categories [29]

3. RESEARCH GAP

Most of the prior work in the identification of emotions has been done through facial recognitions, but the above mentioned limitations make facial recognition not accurate and create ethical issues. EEG method brings out the best results but this is an uncomfortable method. To avoid these limitations this study will focus on art work a child has drawn and the ERC the teachers are filling to make it reliable. The output will be produced to explain why the emotions was chosen. A color clustering will also help to identify the emotion and color connectivity.

research	input	Cultural context	Explanation	Background context	Labeling emotions
[22]	drawing	✗	✗	✗	✓
[23]	drawing	✗	✗	✗	✓
[24]	drawing	✗	✗	✗	✓
[25]	Drawing	✓	✗	✗	✓
[26]	drawing	✗	✗	✗	✓
[27]	Drawing	✗	✗	✗	✓
My app	Art + ERC	✓ Sri Lankan colors	✓ color clustering + ERC + art labeling	✓ ERC	✓ Happy, Sad, Anger

Table 3.1: Prior works in drawings and emotions

The novelty of this project lies in its ability to incorporate both behavior and artworks to identify emotions. In prior work, drawings were only labeled, making them harder to trust the emotion produced by the system. By incorporating emotion regulation checklist, the system makes the users easier to identify why that emotion was taken. Using knowledge in cultural meanings of the colors, makes it a cultural relevant method as well. Emotion Regulation Checklist (ERC) is used widely in research and the accuracy of the ERC in prior researches make it a more reliable source.

4. RESEARCH PROBLEM

Analyzing emotions in children are crucial for early childhood development [22]. When a small child cannot talk about their feeling, they start to lock them self, adopts an attitude of withdrawal [7]. After wards these emotions starts to become uncontrollable reactions. By neglecting the child's emotional needs will negatively impact early childhood development, especially for children who cannot express their feelings clearly[21].

Neglecting a child's emotional distress might cause them to not trust people, effect their relationships and studies. This is especially through when it comes to foster children, children who just lost a parent, refugee, special needs children, adopted children, children who are coming from a violent family background.

Many studies have been conducted to identify the emotions of children, widely used method for this is facial expression but this method has its limitations due to not being able to capture the hidden emotions. The most reliable method of identifying emotions are through EEG but these also fails when it comes to accessibility and comfortableness.

Therefore the research problem is to find an emotion identification system where there is no digital addition for children, respect the privacy of the child and avoid the limitations that are in facial expression applications. A method that would also be relevant to the Sri Lankan culture. This should also address the transparency issue that the prior emotion recognition applications had.

5. OBJECTIVES

Main objective

Help the parent and teacher monitor the child's emotional intelligence development and find whether any critical situation has occurred.

Specific objective

- Identify the emotion the child is going through using art work and ERC given by the teacher.
- Cluster colors in images to see what the most used colors by the student are
- Explain how the emotions were taken
- Calculating and classifying the ERC results to check the negativity labeling

6. METHODOLOGY

The study will involve children aged 4–6 years from selected preschools in Urban areas in Sri Lanka. Participation will require teacher agreement. Pre school teachers and an expert in the industry will be engaged for emotional labeling and interpretation.

6.1 Requirements Gathering

Before the start of the project we visited HummingBird Learning Center, Busy Bees, Malabe Christine Alexandria preschool in western province, to see how they analyzed the emotional intelligence in students. We also met with our external supervisor Dr. Naren Deepan to get the psychological knowledge for our research

6.2 Data Collection

Art work data

- Each child will be invited to produce free-drawing and guided-drawing artworks in a classroom setting.
- Drawing sessions will be conducted in familiar environments to minimize performance anxiety.
- Artworks will be digitally scanned in high resolution for computational analysis.

Behavioral Observations

- Teacher observations will be recorded using a structured checklist based on the Emotion Regulation Checklist (ERC), focusing on Lability/Negativity and Emotion Regulation subscales.
- Teacher qualifications, years of experience, and familiarity with the child will also be documented to assess observer reliability.

6.3Labeling of Emotional States

- Teachers will provide initial emotional state labels (e.g., happy, sad, angry, anxious, neutral) based on classroom interactions.
- An industry expert will review and validate these labels to ensure psychological accuracy and reduce bias.

Preschool validity and quality

- It rule initiated by the government, that two teachers should be available for 20 children in a preschool.
- All preschool teachers must be qualified and have to be qualified, they should have completed, early childhood education course and NVQ levels.
- The observations that are done by the preschool teachers about the students are approved by the teacher in charge and also the head mistress of the preschool.

The child's anomality will be kept to ensure the ethical aspect. The child's participation will be entirely voluntary and the ethical approval will be obtained from the relevant institution.

System Overview

The inputs will be uploaded into the system and the process will drawings will be classified into happy, sad, angry emotions and the Emotion Regulation Checklist will help build the accuracy more with negativity/lability subscale. The system overview is shown in the diagram below,

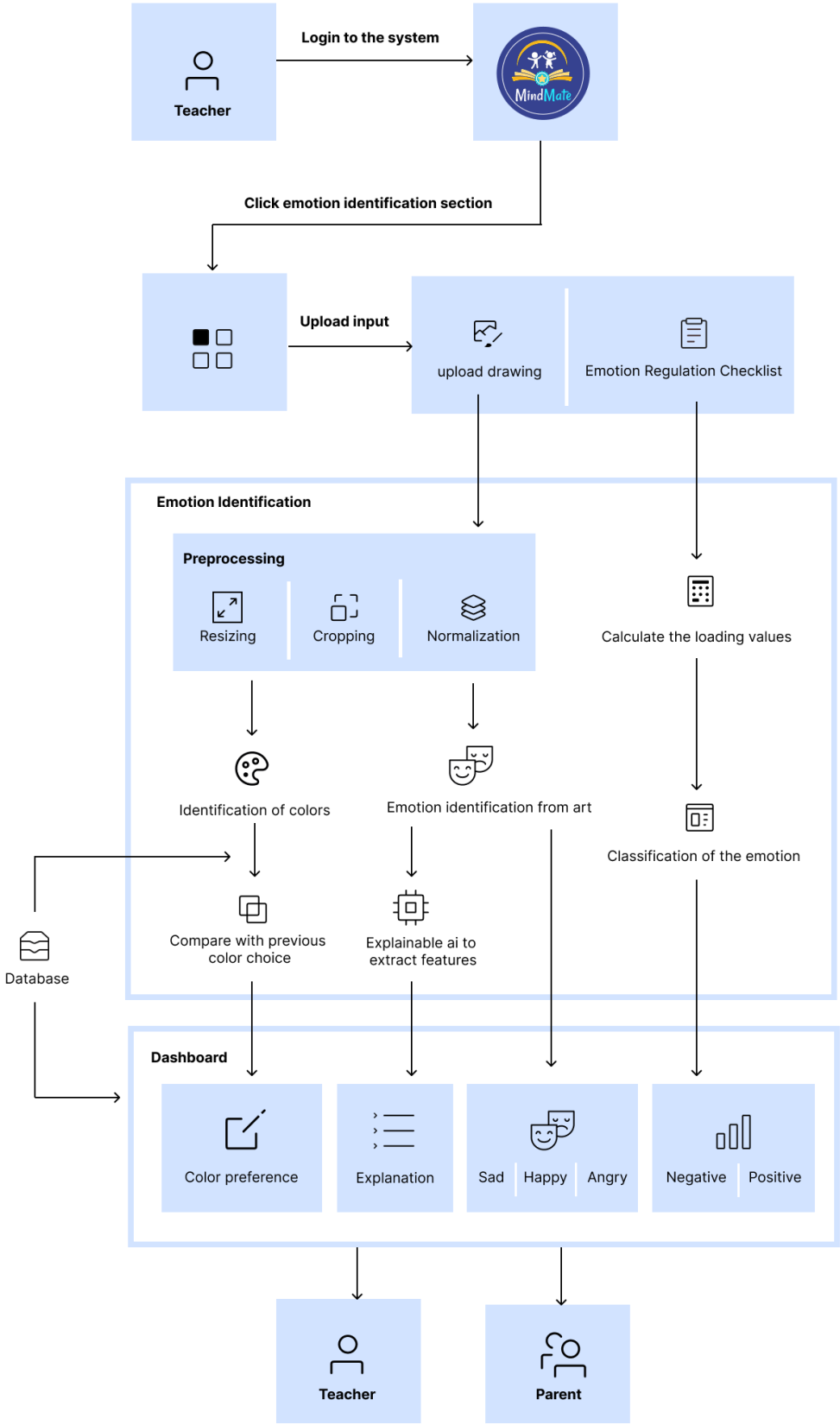


Figure 6.3 1: Component Diagram

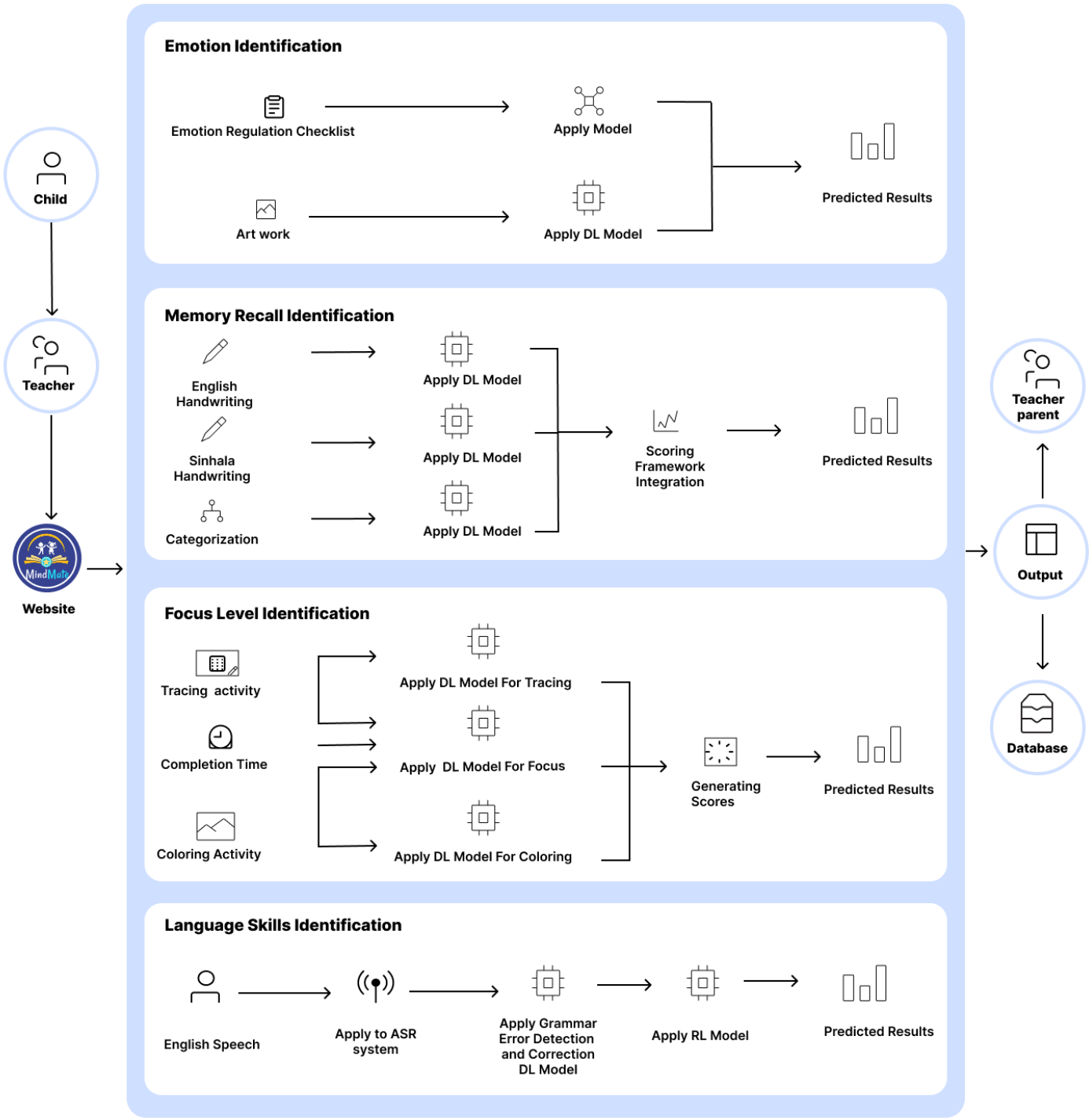


Figure 6.3 2: System Diagram

6.4 Tools and Technologies

Deep Learning

Since the data is small a deep learning method will be used to train the data. Deep learning is also able to capture the complex feature patterns making it an ideal method. The transfer learning ability of the deep learning technique helps with the small dataset issue.

Image augmentation

Image processing techniques, such as image augmentation will be used to increase the dataset, which will help the model train more. Which will help the model learn features.

Explainable AI

Deep learning is a method that can learn without manually extracting the data but this is a black box method where the user cannot see which features where mostly used to label the data. To bring the transparency and the trust worthiness we can use an explainable ai method. This will be used extract features that were used in the drawing.

Image preprocessing

Resizing of the image will be used to maintain the dataset quality making it easier for the model. The image will be normalized to 0-1.

Database - Postgres SQL

Languages – Python, react js

This will be a web based application. The main ide used will be visual studio code. The models will be trained in a Jupiter environment.

7. PROJECT REQUIREMENT

7.1 Functional Requirement

- Ability to upload the drawings to the system
- Ability to answer the Emotional Regulation Checklist
- Ability to understand the colors that were used
- Considering the emotion based on the colors based on the provided knowledge
- Extract the features from the drawing and come up with the emotional state of the drawing
- Calculate the scoring of the Emotional Regulation Checklist and label negativity
- Store the details in the database

7.2 Non Functional Requirement

- Reliable
  The system should be trust worthy.

- Accuracy
  The system should be able to produce accurate results since this is related directly to children

- Transparency
  System should be able to explain why it classified that emotion. Making the user easier to understand and trust the system.

- User friendly
  It should also be user friendly because non-technical people will be accessing the system

- Secured
  Since this will be storing the child's information, only the teachers and the parent should be able to view the data.

7.3 **Test Cases**

Test Case ID	Art	ERC Negativity/Lability	Expected Output
TC01	Sad	High	Frequent mood swings, irritability, and emotional instability; sadness in art reflects persistent distress and poor coping skills
TC02	Angry	High	Strong indicators of frustration and impulsive reactions; anger in art mirrors actual behavioral regulation struggles.
TC03	Happy	High	Outwardly positive art but frequent underlying mood shifts; happiness may mask regulation issues.
TC04	Sad	Low	Stable emotional baseline but artwork reflects temporary sadness (possibly situational or creative choice).
TC05	Angry	Low	Anger in art is less about emotional instability and more possibly thematic, storytelling, or creative exploration.
TC06	Happy	Low	Stable positive affect; happy artwork reflects genuine emotional well-being.
TC07	Sad	Moderate	Sadness may be tied to occasional negative mood states; emotional regulation is sometimes effective but not consistent.
TC07	Angry	Moderate	Anger expressed visually may stem from situational frustration; child can regulate mood in some contexts but struggles in others.
TC07	Happy	Moderate	Generally positive mood with some moments of frustration or emotional fluctuation.

Table 8.3 1: Testing cases

7.4 Gnatt Chart

Task Name	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Research topic selection												
Topic assesement												
Project charter												
Study on research area												
Project proposal report												
System design and planning												
Implementation of functions												
Integration level 1												
Testing level 1												
Progress presentation 1												
Check List 1												
Prepare research paper												
Research paper												
Implementation of functions												
Testing level 2												
Progress presentation 2												
Check List 2												
Final report												
Final presentation												

Figure 7.4 1: Gnatt chart

7.5 Work break down

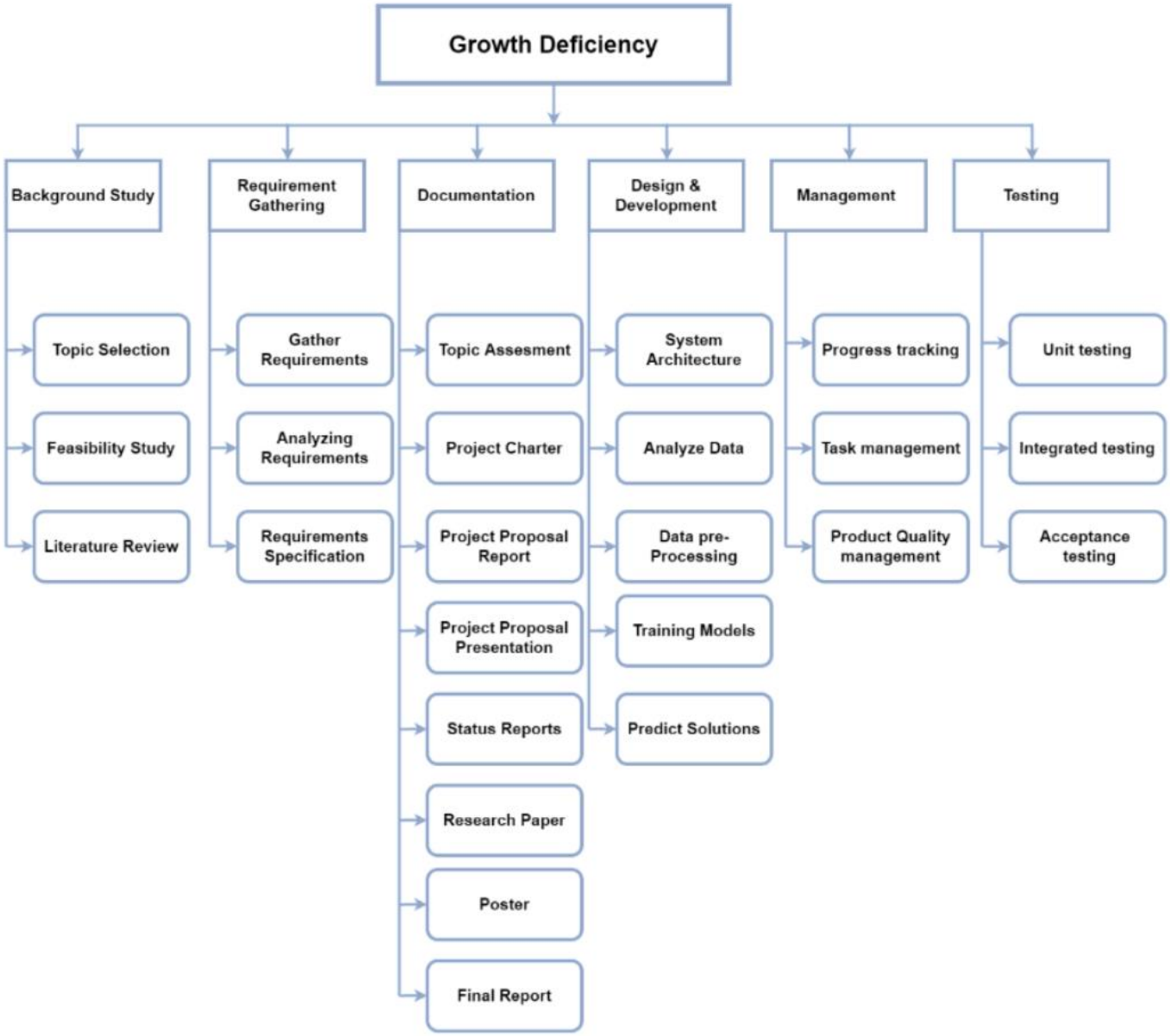


Table 7.3 2: work break down

8. COMMERCIALIZATION

8.1 Customer value

Through the survey results we found out, 85% of the people would like to see an AI tool to monitor the emotional intelligence. They also highlighted how important emotional intelligence monitoring is for the early childhood development. The target audience for this application will be parents and teachers. Considering these results below are some of the steps that will be considered to commercialize the application.

8.2 Competitive analysis

Similar applications on emotional intelligence without AI models has an average price range of $9.99 monthly. Applications that use AI models has an average price of $20 monthly subscription.

How our application differ from the competitors?

Exiting applications	Our application
Gamified applications that increase digital addictions of the children	Uploading the work the child has done to reduce digital addiction
Facial expression capturing to identify emotions	Does not capture or store children's facial expressions or biometric identifiers
Use activities to improve child specific skill area.	Use activities not only to improve their skills but also to get insights on emotional intelligence based on those activities.
Limited parental involvement or communication	Facilitates and enhances parent-teacher interaction and collaboration
Focus more on cognitive, thinking, and emotions with reward-based motivation	Provides meaningful insights and personalized recommendations to parents and teachers based on the child's activity outputs
Primarily rely on digital input (typing, drag-and-drop, touch-based tasks)	Ensures that no direct digital input is involved in children's activities.
Generic design without consideration for cultural or linguistic diversity	Incorporates both Sinhala and English, ensuring cultural and linguistic appropriateness.

Figure 8.2 1: Our application vs existing applications

User benefits of our applications

- Monitor child's emotional intelligence using multiple methods
- Builds emotional intelligence at an early age, which supports learning, social skills, and resilience.
- Helps teachers identify at-risk students early (e.g., children showing signs of sadness, anger, withdrawal)
- Teachers and parents both can help children to improve suggested activity skills based on dashboard feedback.

8.3 Cost analysis

On average a web hosting platform will cost $7 with postgesql database support (eg: Hostinger). This will be the rough hosting cost of our application.

8.4 Pricing plan

Considering the above factors this application will have a one month free trial period, which will be used to gain customers into the application. After the one month trial, the user will have to pay $15 monthly subscription.

Standard plan ($15) – One subscription per student.

Family plan ($30) – One subscription per three students.

School plan ($40) – Class registration ability, one subscription per 20 students.

References

[1] V. Pawar, A. Vhatkar, P. Chavan, S. Gawankar, and S. Nair, “The future of emotional engineering: Integrating generative AI and emotional intelligence,” in *Proc. 2024 8th Int. Conf. Computing, Communication, Control and Automation (ICCUBEA)*, Pune, India, Aug. 2024, pp. 1–6, doi: 10.1109/ICCUBEA61740.2024.10775105.

[2] I. Coronado-Maldonado and M.-D. Benítez-Márquez, “Emotional intelligence, leadership, and work teams: A hybrid literature review,” *Heliyon*, vol. 9, no. 10, art. e20356, Oct. 2023, doi: 10.1016/j.heliyon.2023.e20356.

[3] J. Ebert et al., "Teaching Emotion and Creativity Skills to Preschoolers Through a Child Workshop," *The International Journal of Creativity & Problem Solving*, vol. 25, no. 2, pp. 23–35, 2015.

[4] Zhanna Iksanova, “A new method of developing emotional intelligence in preschool children through interactive digital facial expression recognition technology”, *in Proc. 2022 4th Int. Conf. Neurotechnologies and Neurointerfaces (CNN)* Kaliningrad, Russian Sep 2022, pp. 1-6, doi: 10.1109/CNN56452.2022.9912512

[5] A. L. Ruba and S. D. Pollak, "The Development of Emotion Reasoning in Infancy and Early Childhood," *Annual Review of Developmental Psychology*, vol. 2, pp. 503–531, 2020, doi: 10.1146/annurev-devpsych-060320-102556.

[6] S. Richard et al., "The development of specific emotion comprehension components in 1285 preschool children," *Scientific Reports*, vol. 12, no. 1, p. 19782, 2022, doi: 10.1038/s41598-022-22525-4.

[7] T. Kim et al., "Application of Deep Learning in Art Therapy," *International Journal of Machine Learning and Computing*, vol. 11, no. 6, pp. 496–500, 2021.

[8] H. Gomi et al., "Cross-Linguistic Study on Affective Impression and Language for Visual Art Using Neural Speaker," in *2022 10th International Conference on Affective Computing and Intelligent Interaction (ACII)*, 2022, pp. 1–6, doi: 10.1109/ACII55700.2022.9953880.

[9] K. A. R. Bucag, I. J. P. Casidsid, J. T. S. Enicola, and A. L. Ilao, “ADOMATH: An Android mobile game application for preschoolers,” in *Proc. 2024 22nd Int. Conf. ICT and Knowledge Engineering (ICT&KE)*, Bangkok, Thailand, Nov. 2024, pp. 1–6, doi: 10.1109/ICTKE62841.2024.10787159.

[10] E. Sudarmilah, A. Susanto, R. Ferdiana, and N. Ramdhani, “Developing a game for preschoolers: What character, emotion and reward will tend to hack preschoolers?,” in *Proc. 2015 Int. Conf. Data and Software Engineering (ICoDSE)*, Yogyakarta, Indonesia, Nov. 2015, pp. 1–6, doi: 10.1109/ICODSE.2015.7436977.

[11] M. K. Mokhtar, F. Mohamed, M. S. Sunar, M. A. M. Arshad, and M. K. M. Sidik, “Development of mobile-based augmented reality colouring for preschool learning,” in *Proc. 2018 IEEE Conf. e-Learning, e-Management and e-Services (IC3e)*, Langkawi, Malaysia, Nov. 2018, pp. 1–6, doi: 10.1109/IC3e.2018.8632639.

[12] B. Parihar, S. Chhoriya, and R. Sangle, “Go-Brainous: An AI based educational app for kids,” in *Proc. 2023 11th Int. Conf. Emerging Trends in Engineering & Technology – Signal and Information Processing (ICETET-SIP)*, Nagpur, India, Apr. 2023, pp. 1–6, doi: 10.1109/ICETET-SIP58143.2023.10151633.

[13] R. Ramos et al., "Emotion Analysis in Children's Drawings Using Machine Learning Techniques," *Research in Computing Science*, 2024.

[14] Ş. Danışman, E. D. İman, Z. A. Demircan, and D. Yaya, "Examining the psychometric properties of the Emotional Regulation Checklist in 4- and 5-year-old preschoolers," *Electronic Journal of Research in Educational Psychology*, vol. 14, no. 3, pp. 534–556, 2016, doi: 10.14204/ejrep.40.15124.

[15] A. H. Reis et al., "Emotion Regulation Checklist (ERC): Preliminary Studies of Cross-Cultural Adaptation and Validation for Use in Brazil," *Temas em Psicologia*, vol. 24, no. 1, pp. 77–96, 2016.

[16] C. Santos et al., "Factorial structure, measurement invariance and reliability of the Emotion Regulation Checklist (ERC) in a sample of Portuguese parents," *Análise Psicológica*, vol. 42, no. 2, pp. 197–207, 2024, doi: 10.14417/ap.2032.

[17] A. P. N. Sari, B. H. R. G. P. F. Adha, and E. S. D. Wulandari, "Theory of Mind, Roles, and the Development of Emotion Regulation in Early Childhood," *JPUD - Jurnal Pendidikan Usia Dini*, vol. 18, no. 1, pp. 115–125, 2024, doi: 10.21009/10.21009/JPUD.181.09.

[18] C. J. Cress, L. Synhorst, M. H. Epstein, and E. Allen, "Confirmatory Factor Analysis of the Preschool Behavioral and Emotional Rating Scale (PreBERS) With Preschool Children With Disabilities," *Assessment for Effective Intervention*, vol. 37, no. 4, pp. 203–211, 2012, doi: 10.1177/1534508411433499.

[19] M. H. Epstein, L. L. Synhorst, and C. J. Allen, "Development and Standardization of a Test to Measure the Emotional and Behavioral Strengths of Preschool Children," *Journal of Emotional and Behavioral Disorders*, vol. 17, no. 1, pp. 29–37, 2009, doi: 10.1177/1063426608319223.

[20] M. Ebert et al., "Creativity, Emotion, and Art: Development and Initial Evaluation of a Workshop for Professional Adults," *The International Journal of Creativity & Problem Solving*, vol. 25, no. 2, pp. 47–59, 2015.

[21] A. Alshahrani, M. M. Almatrafi, and J. I. Mustafa, "A Children's Psychological and Mental Health Detection Model by Drawing Analysis based on Computer Vision and Deep Learning," *Engineering, Technology & Applied Science Research*, vol. 14, no. 4, pp. 15533–15540, 2024.

[22] R. Ramos et al., "Emotion Analysis in Children's Drawings Using Machine Learning Techniques," *Research in Computing Science*, 2024.

[23] B. Beltzung et al., "Deep learning for studying drawing behaviour," *Frontiers in Psychology*, vol. 14, 2023, doi: 10.3389/fpsyg.2023.992541.

[24] A. Alshahrani, M. M. Almatrafi, and J. I. Mustafa, "A Children's Psychological and Mental Health Detection Model by Drawing Analysis based on Computer Vision and Deep Learning," *Engineering, Technology & Applied Science Research*, vol. 14, no. 4, pp. 15533–15540, 2024.

[25] H. Gomi et al., "Cross-Linguistic Study on Affective Impression and Language for Visual Art Using Neural Speaker," in *2022 10th International Conference on Affective Computing and Intelligent Interaction (ACII)*, 2022, pp. 1–6, doi: 10.1109/ACII55700.2022.9953880.

[26] Y. Wang, "Sentiment analysis of ethnic artworks using deep learning," *Applied Mathematics and Nonlinear Sciences*, 2024.

[27] Z. Dang, "Sentiment analysis of art and design works using deep learning," *Applied Mathematics and Nonlinear Sciences*, vol. 9, no. 1, pp. 1–16, 2024.

[28] A. J. Elliot, "Color and psychological functioning: a review of theoretical and empirical work," *Front. Psychol.*, vol. 6, p. 368, 2015, doi:10.3389/fpsyg.2015.00368.

[29] D. Jonauskaite and C. Mohr, "Do we feel colours? A systematic review of 128 years of psychological research linking colours and emotions," *Psychonomic Bulletin & Review*, 2025, doi: 10.3758/s13423-024-02615-z.

[30] A. A. Hettiarachchi and N. De Silva, "Colour associated emotional and behavioural responses: A study on the associations emerged via imagination," *Built - Environment - Sri Lanka*, vol. 11, no. 1, pp. 21–27, 2012.

Appendix 1: Validation letter of the art work

CHRISTINE ALEXANDRIA ACTIVITY SCHOOL

Spectrum Studio – The Therapeutic art Centre

Malabe. 0774089071

To The Research Supervisor.

This is to certify that **R. E. Chathurika**, a student conducting the research titled *"AI-Based Early Childhood Emotional Intelligence Monitoring Platform in Urban Areas"*, has collected a large number of children's drawings for her study.

On her request, these drawings were categorized and validated into three emotional states: **happy, sad, and angry**. This categorization was done solely for the purpose of supporting her research and model training.

Validation / Authorization

W. C. Niroshani
Attorney -at -Law
BSc (Psychology), MSc (Psychology), MEd
Education and Developmental Psychologist & Therapist
Certified Art Therapist

Thank you.

Yours sincerely,

W. C. Niroshani
Education and Developmental Psychologist & Therapist
Certified Art Therapist

Appendix 2: ERC questions

ERC items	EFA 24 items		EFA 23 items			
	L/N	ER	L/N	ER	r*	α**
20: Impulsivity	.842	-	.833	-	.71	.83
14: Anger at limits	.820	-	.805	-	.73	.83
08: Outbursts of anger	.800	-	.803	-	.74	.83
13: Outbursts of enthusiasm	.707	-	.737	-	.65	.83
22: Intrusive enthusiasm	.698	-	.727	-	.62	.84
17: Over-ebthusiastic	.695	-	.771	-	.56	.84
24: Negative emotions at invitations to play	.660	-	.609	-	.57	.84
06: Frustration	.642	-	.627	-	.64	.83
02: Mood swings	.602	-	.588	-	.67	.83
10: Pleased to see others suffer	.442	-	.449	-	.52	.84
12: Crying and clinging to adults	.343	-	.327	-	.40	.85
09: Able to delay gratification	-.307	-	-.329	-	.51	.85
11: Excitation control	-.306	-	-.331	-	.46	.85
01: Happiness	-	.755	-	.782	.58	.72
03: Positive response to adult approaches	-	.733	-	.763	.65	.71
15: Talk about negative emotional states	.301	.680	.320	.695	.58	.73
18: Apathetic mood	-	-.656	-	-.686	.54	.73
07: Positive response to peer approaches	-	.628	-	.643	.59	.72
21: Empathy	-	.622	-	.558	.50	.74
05: Recovering from negative emotions	-	.512	-	.503	.59	.73
16: Sadness and apathy	-	-.459	-	-.489	.49	.74
19: Negative response to peer approaches	.359	-.410	.331	-.442	.51	.73
04: Can switch well of activities	-	.385	-	.352	.54	.74
23: Appropriate negative emotions to the demand	.330	.368				
Cronbach's Alpha	.848	.749	.848	.749		
Inter-factors correlation	-.459		-.490			
Common variance explained	54.45%		56.95%			

Note. ERC = Emotion Regulation Checklist; EFA = Exploratory Factor Analysis. L/N = Factor 1 "Lability/Negativity"; ER = Factor 2 "Emotional Regulation"; * Item-total correlation; ** Cronbach's Alpha if item deleted; "-" Factorial loading less than .300.

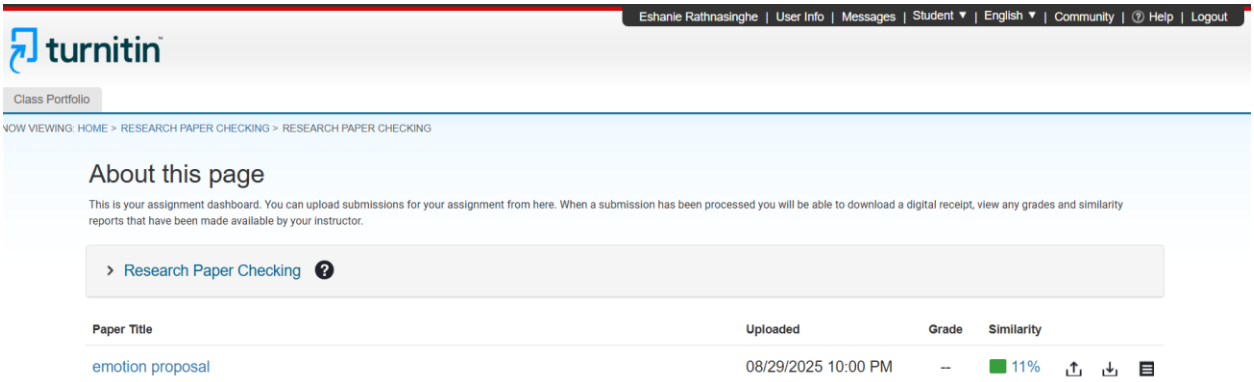
Appendix 3: ERC question loadings

Standardized factor loadings for two-factor model CFA

Subscale	Item	λ
Emotion Regulation	Item 1	.41
	Item 3	.49
	Item 7	.65
	Item 15	.46
	Item 16*	.26
	Item 18*	.26
	Item 21	.48
	Item 23	-
Lability/Negativity	Item 2	.59
	Item 4*	-
	Item 5*	.22
	Item 6	.62
	Item 8	.72
	Item9*	.24
	item 10	.36
	Item 11*	-
	Item 13	.65
	Item 14	.70
	Item 17	.36
	Item 19	.34
	Item 20	.56
	Item 22	.54
	Item 24	.32

Appendix 4: MindMate Logo

Appendix 5: Plagiarism Check



emotion proposal

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